

Cookbook Memory

AI Agent Memory Harness — persistent memory for long-running coding agents

Codename **Cookbook Memory** · Team of 4 (Keith · Ken · Brent · Scott B.) · Two-week sprint · Validated on five public benchmarks

1 The problem

There is **no standard, model-agnostic layer** that decides **what to remember, where to store it, how to retrieve it fast**, and **how to keep it consistent over time** for long-running agents.

DECISION	WHAT IT MEANS
What	signal vs. noise
Where	the right store
Fast	no context flood
Consistent	dedup, no conflicts

2 Hypothesis & success criteria

Hypothesis. With the memory harness, **Haiku can close the gap to Opus 4.8** (which runs without memory) on established memory benchmarks.

Success criterion. On **at least two of the five** benchmarks, *Haiku + harness* beats the *Opus 4.8 no-memory* baseline across the four metrics, **without** blowing the efficiency budget (memory adds **< ~10%** context-token overhead).

METRIC	DEFINITION
Recency	Is the freshest relevant memory ranked first?
Efficiency	Tokens per retrieval; target under ~10% overhead.
Relevancy	Retrieved items actually relate to the query.
Accuracy	Task correctness, memory on vs. off.

3 Technical approach

A **pluggable memory harness**: four modules over three indexed storage backends, wired into the open-source **OpenCode** agent and measured on public benchmarks.

Four modules

- **Persistence layer (write).** Decides what/when/where to save; tags each item (timestamp, relevancy, session); versioning + dedup hints.
- **Intelligent router (dispatch).** Classifies each query and hits the *single best* backend. Rule-based, then learned.
- **Retrieval orchestrator (read).** Unified API: route, rank by `recency × relevancy`, de-dup, return a tight context.
- **Dreaming component (async).** Runs **while other agents sleep**: de-dup, resolve contradictions, session governance (must-know / must-do / blacklist), retention + pruning.

Three storage backends (each indexed)

BACKEND	BEST FOR	INDEX
Markdown + YAML	Literal recall	Inverted keyword map
SQLite + vectors	Semantic search	Dense ANN (HNSW / FAISS)
Graph store	Relationships & conflicts	Typed traversal

Evaluation

Public benchmarks only — no home-grown evals. Per benchmark, a controlled grid runs Haiku + harness vs. Haiku, Opus 4.8 and Sonnet — prompts and task order fixed, every trajectory logged.

4 Scope

In scope

- The memory harness: all four modules, end to end.
- Three indexed storage adapters behind one interface.
- The intelligent router (rule-based; learned-classifier hook).
- The async dreaming component: dedup, conflict resolution, governance, retention.
- OpenCode integration for memory write/read on each step.
- Eval harness over five benchmarks; baselines for Haiku, Opus 4.8, Sonnet.
- Metric definitions, a reproducible protocol, results dashboard.

Out of scope (non-goals)

- Building our own benchmark or dataset — we use public ones.
- Training or fine-tuning any model.
- Production hardening: multi-tenant infra, auth, SLAs, polished UI.
- Distributed / scaled deployment — single-node is enough for the sprint.
- Validating non-coding agents — design generalizes, but not tested now.

5 Team & ownership — who owns what

Four parallel workstreams, locked behind a shared storage interface **frozen on Day 3**.

The team: **P1 Keith** (harness, OpenCode) · **P2 Ken** (benchmark) · **P3 Brent** (store, retrieve) · **P4 Scott B.** (dreaming).

AREA / DELIVERABLE	OWNER	SUPPORTING
Storage interface & memory-item schema	P1 · Keith	P3
Persistence layer (write path, tagging, versioning)	P1 · Keith	—
Retrieval orchestrator (rank · dedup · return)	P1 · Keith	P3
OpenCode integration (write/read on each step)	P1 · Keith	—
Three storage backends + per-backend indexes	P3 · Brent	P1
Intelligent router (rules → learned)	P3 · Brent	P1
Backend performance testing	P3 · Brent	P2
Dreaming worker (dedup, conflict, governance, retention)	P4 · Scott B.	P1, P3
Memory semantics ("what is good memory")	P4 · Scott B.	P2
Datasets, loaders & trajectory logging	P2 · Ken	P4
Metric defs, shared run harness & cost gates	P2 · Ken	P4
Results + cost dashboard, stats, aggregation	P2 · Ken	—
Run SWE-Bench-CL eval (own key)	P1 · Keith	—
Run LongMemEval eval (own key)	P2 · Ken	—
Run SWE-ContextBench eval (own key)	P3 · Brent	—
Run MemoryAgentBench eval (own key)	P4 · Scott B.	—
Run ContextBench eval (own key)	P3 · Brent	—
Final end-to-end integration	All	—

- **P1 · Keith — Harness Architecture & OpenCode Integration.** Critical path — the contracts everyone builds against.
- **P2 · Ken — Evaluation Infrastructure & Coordination.** Shared runner, metrics, dashboard — and captains one benchmark.
- **P3 · Brent — Storage Implementation & Router.** Three fast backends + the dispatch layer; captains the two retrieval benchmarks.
- **P4 · Scott B. — Dreaming Component & Memory Governance.** The offline engine that keeps memory clean.

Dividing the eval runs (API cost & time)

~20 runs (5 benchmarks × 4 configs) is too much cost and time for one person on one key. Each teammate **captains** the benchmark(s) that stress their own component, on their **own API budget**.

BENCHMARK	CAPTAIN	WHY THEM
SWE-Bench-CL	Keith (P1)	Drives the coding agent end-to-end — his OpenCode wheelhouse.
LongMemEval	Ken (P2)	Eval lead; recency / temporal reasoning.
SWE-ContextBench	Brent (P3)	Cross-task context reuse exercises his retrieval + router.
MemoryAgentBench	Scott B. (P4)	Tests conflict resolution — his dreaming component.
ContextBench	Brent (P3)	Retrieval-quality (gold contexts) — the retrieval/router he owns.

Ken owns the shared runner — `run(benchmark, model, memory) → metrics` — and aggregates all results. Controls: sharded keys (one per captain); cheapest-first + early-exit (Opus last, confirm-only); dev-slice → full; cache + resume; hard per-benchmark budget gates; no-memory baselines in week 1.

6 Two-week timeline

Person 1 · Keith — Harness Architecture & OpenCode Integration

- **Week 1:** Three-module abstraction & storage interface. Persistence layer (write, versioning, tagging). Instrument OpenCode to write memory each step.
- **Week 2:** Retrieval orchestrator (rank → dedup → return); wire in Brent's router; end-to-end task. **Captain the SWE-Bench-CL runs.**

Person 2 · Ken — Evaluation Infrastructure & Coordination

- **Week 1:** Parse all five datasets; loaders + trajectory logging (with Scott B.). Define the metrics; build the shared run harness + cost tracker.
- **Week 2:** **Captain the LongMemEval runs.** Aggregate every captain's results; stats + results & cost dashboard; document the protocol.

Person 3 · Brent — Storage Implementation & Router

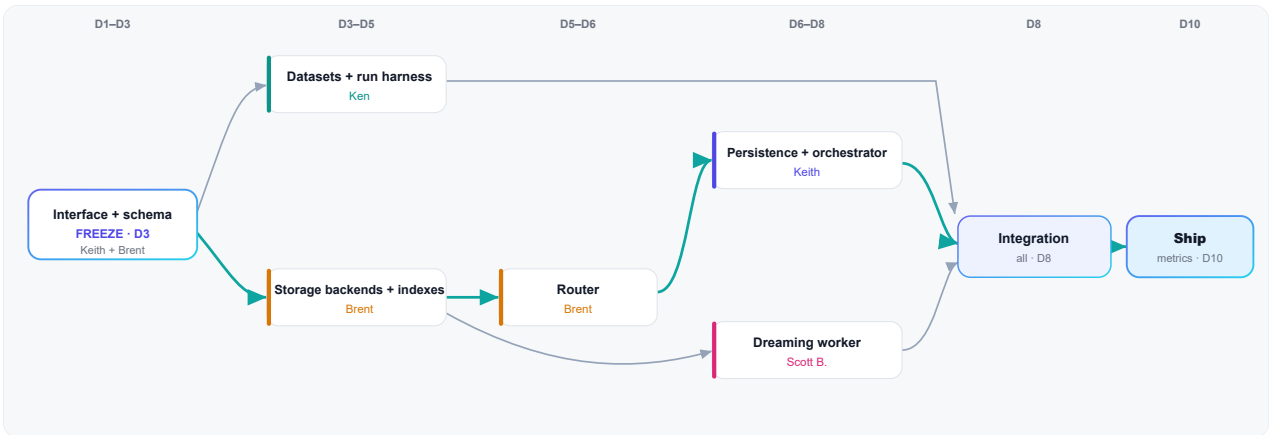
- **Week 1:** SQLite + vector pipeline (HNSW/FAISS). Markdown store + inverted index. Graph store (Neo4j) + traversal index.
- **Week 2:** Adapters behind Keith's interface; implement the router; performance-test each backend. **Captain the SWE-ContextBench & ContextBench runs.**

Person 4 · Scott B. — Dreaming Component & Memory Governance

- **Week 1:** Dedup (exact / semantic / near-dup). Conflict detection + reconciliation. Session-filter & blacklist. Co-build trajectory logging with Ken.
- **Week 2:** Async scheduler that runs while agents sleep; must-know / must-do extraction; retention & pruning; observability; integration with Keith & Brent. **Captain the MemoryAgentBench runs.**

7 Dependency map & critical path

The Day-3 interface freeze unblocks every stream; the bold chain is the critical path to ship.



Teal = critical path (freeze → backends → router → orchestrator → integration → ship). Grey = supporting dependencies.

- **Keith + Brent (D1–D3):** co-author and freeze the storage interface + schema, so persistence and adapters share one contract.
- **Brent → Keith:** the router lands before the orchestrator can route in week 2.
- **Ken → all (by D5):** datasets, logging and the shared run harness ready, so baselines can start ~D4–D6.
- **Keith + Brent → Scott B.:** the dreaming worker integrates once real storage exists (D8+).
- **Sharded keys:** each captain on a separate API budget — baselines week 1, treatment week 2.

Top risk is semantic, not structural: defining *what counts as "good memory."* Ken and Scott B. must align on memory semantics in week 1, or the harness stores everything and retrieves nothing useful.

8 Milestones

DAY	MILESTONE
D3	Contract freeze — storage interface + memory-item schema locked; metric definitions agreed.
D5	End of week 1 — three backends read/write; run harness + loaders ready; no-memory baselines started on sharded keys.
D8	Integration start — router + orchestrator connected; dreaming worker running against real stores.
D10	Ship — all benchmark shards complete (Haiku + harness); four metrics aggregated vs. baselines.

9 Benchmarks & metrics

BENCHMARK	PRIMARY METRICS	REFERENCE
MemoryAgentBench	Relevancy, Accuracy, conflict resolution	arXiv 2507.05257
LongMemEval	Recency, Accuracy, temporal reasoning	arXiv 2410.10813
SWE-ContextBench	Efficiency, Relevancy, Accuracy	arXiv 2602.08316
SWE-Bench-CL	Relevancy, Accuracy, continual learning	arXiv 2507.00014
ContextBench	Relevancy, Efficiency, retrieval recall/precision	arXiv 2602.05892

Full descriptions, dataset links and the metric-mapping matrix live on the project site's **Benchmarks** page.

10 Collaboration & deliverables

Four developers, one repo, no conflicts: **prd.md** / **architecture.md** / **plan.md** are the contracts; GitHub Flow + CODEOWNERS + branch protection enforce one-owner-per-directory and all-owner sign-off to change a frozen interface. See `CONTRIBUTING.md`.

- A model-agnostic memory harness (4 modules) integrated with OpenCode.
- Three indexed storage backends behind one interface.
- The async dreaming component with governance & pruning.
- A reproducible eval harness (shared runner + cost gates) + baselines on five benchmarks.
- A results scoreboard testing the hypothesis.
- The project site (GitHub Pages) documenting all of the above.